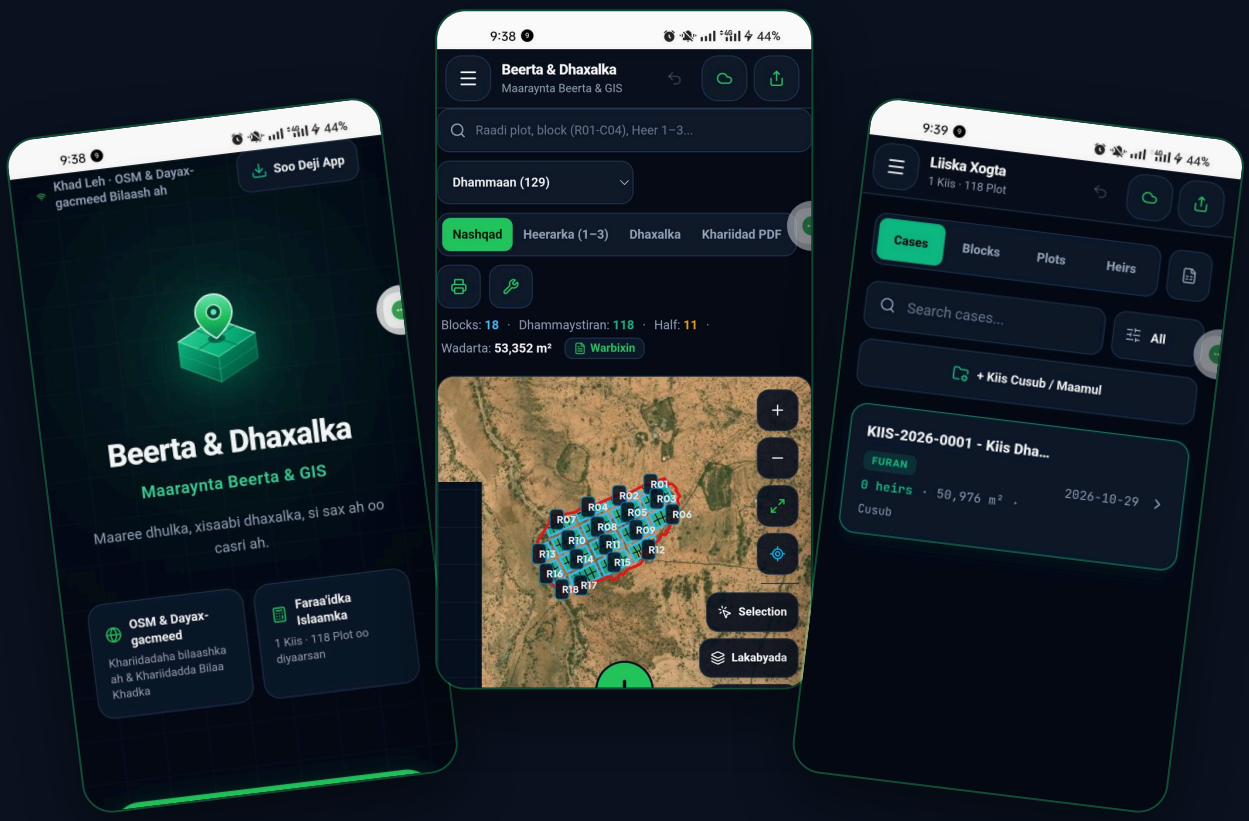
FALANQAYN APP · 30 OKTOOBAR 2026

Beerta & Dhaxalka

Fahamka shaqada appka, khaladaadka la arkay, iyo talooyin design & architecture ah



12	18	10
shaashadood la baaray	khalad & dhibaato	bog oo sawiro talooyin ah

Xaddidaadda shaqada: talooyinka dhammaantood waxay khuseeyaan **muuqaalka, qaab-dhismeedka iyo habka isticmaalka** oo keliya. Habka xisaabinta dhaxalka, xisaabinta plots-ka iyo soo-saarista xogta waa sidoodii — midna lama beddelin.

Bogagga 1–10: Fahamka + khaladaadka

Bogagga 11–20: Sawiro talooyin + architecture + layout tools

Soo koobid: waa maxay appkan?

Waxa aan ka fahmay 12-ka shaashadood

3	129	18	1
qaybood oo muhiim ah (Khariidad · Xisaabi · Xogta)	muuqda (118 plot + 11 half)	block (R01–R18)	kiis (KIIS-2026-0001)

Ujeedada appka

“Beerta & Dhaxalka — Maaraynta Beerta & GIS” waa app mobile ah oo isku xira **labo shaqo oo waaweyn**: (1) **khariidad GIS** oo beer loo qaybiyey blocks iyo plots (R01…R18, P01…P129) lagu muujiyo dayax-gacmeed iyo OSM; (2) **xisaabiyaha dhaxalka Islaamka (Faraa’id)** oo saamiga dhaxlayaasha u beddela plots dhab ah — qoraalka hoose ee menu-ga waxa uu tilmaamayaa Suuradda An-Nisaa (11, 12, 176).

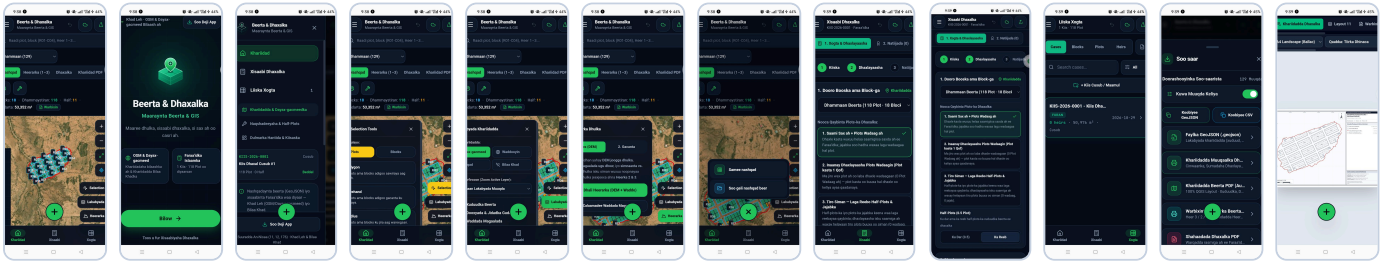
Waxa si fiican u qabta

- **Aqoonsi muuqaal ah oo cad:** mowduuc madow + midab cagaar ah, kaarar leh geesaha duuban — wuxuu u egyahay app casri ah.
- **Shaqo dhammaystiran:** nashqad → xisaab → keydin → soo-saar (GeoJSON, CSV, PDF, Warbixin, Shahaado).
- **Offline:** “Khad Leh” iyo “Bilaa Khad” waxay muujinayaan inuu shaqayn karo khad la’aan.
- **Af-Soomaali:** inta badan interface-ku waa Soomaali, taas oo u habboon isticmaalayaasha beeraha.
- **Qalab GIS oo xooggan:** Polygon, Freehand, Radius, lakabyo, heerarka dhulka (DEM).
- **Layout PDF** oo u eg QGIS (cabir, Waqooyi, legend, xog-ilaha).

Sawirka guud ee khaladaadka

Nooca	Tirada	Tusaale ugu muhiimsan
Muuqaal & layout	9	Panels-ka khariidadda oo bidix ka go’an; FAB oo xogta qarinaya
Xog & is-waafaqsanaan	5	Saddex qiyaasood oo kala duwan oo dhul ah (53,352 m² · 50,976 m² · 9.57 ha)
Luuqad & isticmaal	4	Soomaali iyo Ingiriis oo isku dhafan; icons aan magac lahayn

12-ka shaashadood ee la baaray



#1 #2 #3 #4 #5 #6 #7 #8 #9 #10 #11 #12

Qaabka uu appku u shaqeeyo

Socodka isticmaalaha iyo qaab-dhismeedka xogta ee aan ka fahmay

Socodka guud

1 · Bilow
Splash: “Bilow” ama toos u fur Xisaabiyaha

→

2 · Khariidad
Nashqad beer, xulasho plots/blocks, lakabyo, heerar

→

3 · Xisaabi
Kiiska → Dhaxlayaasha → Natiijada

→

4 · Xogta
Cases · Blocks · Plots · Heirs

→

5 · Soo saar
GeoJSON · CSV · PDF · Warbixin · Shahaado

Qaab-dhismeedka xogta (sida aan u akhriyay)

```
KIIS-2026-0001  (Case / Kiis)
 └─ Beer: xuduud + 18 Block (R01–R18)
     └─ Plots: P01…P129
         ├─ 118 Plot buuxa
         └─  11 Half-Plot (0.5)
 └─ Dhaxlayaasha
     ├─ Marxuum (Nin / Naag)
     ├─ Wiilal · Gabdho · Xaaska
     ├─ Waalidka
     ├─ Ehelka kale
     └─ Deymaha & Hantida
```

Saddex tiir oo appku ku fadhiyo

- **GIS:** khariidad, plots, blocks, heerarka dhulka.
- **Faraa’id:** xisaabinta saamiyada & u qaybinta plots.
- **Diiwaan & Soo-saar:** kiisaska la keydiyey iyo dukumeentiyada rasmiga ah.

Tilmaamo teknikeed (ka soo qaadasho)

- “Soo Deji App” + “Khad Leh” → waxay u egtahay **PWA** oo APK ku duuban.
- Icon-ka daruurta (☁) → xaaladda keydinta/sync.
- Fallback-ka OSM → khariidad tile-based (Leaflet/MapLibre noocaas).

Tani waa fikrad aan ka qaatay shaashadaha; koodka lama arkin.

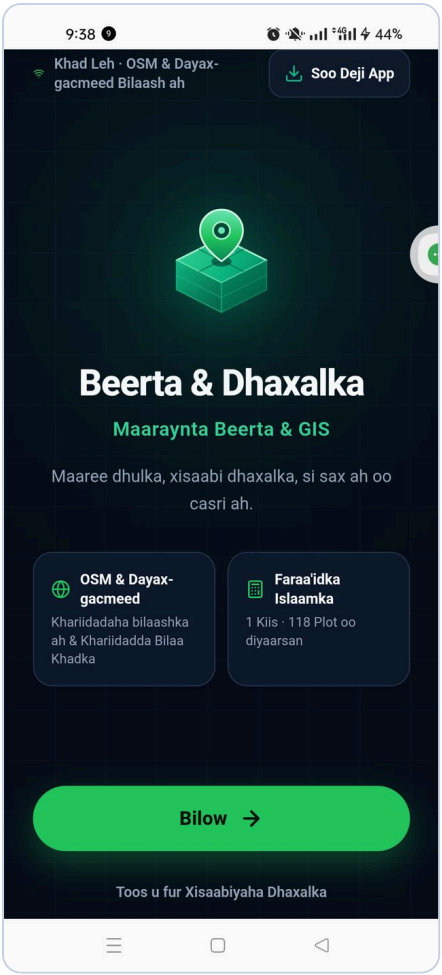
Habka dhex-socodka (navigation)

Saddex meelood ayaa lagu mari karaa: **(a)** bar hoose oo 3 tab leh (Khariidad · Xisaabi · Xogta), **(b)** menu-ga bidix (☰) oo leh qaybo-hoosaad iyo kiiska hadda furan, **(c)** badhanka cagaaran ee “+” (Samee nashqad / Soo geli nashqad beer). Sidoo kale waxaa jira icons sare: ↶ (dib), ☁ (keydin), ⇪ (soo-saar).

Fikradda ugu muhiimsan: kiis (Case) wuxuu ka kooban yahay **beer + dhaxlayaal**. Khariidaddu waxay bixisaa plots-ka; Xisaabigu wuxuu plots-kaas ku qaybiyaa dhaxlayaasha; Xogtu waxay diiwaangelisaa; Soo-saarku wuxuu u beddelaa dukumeenti.

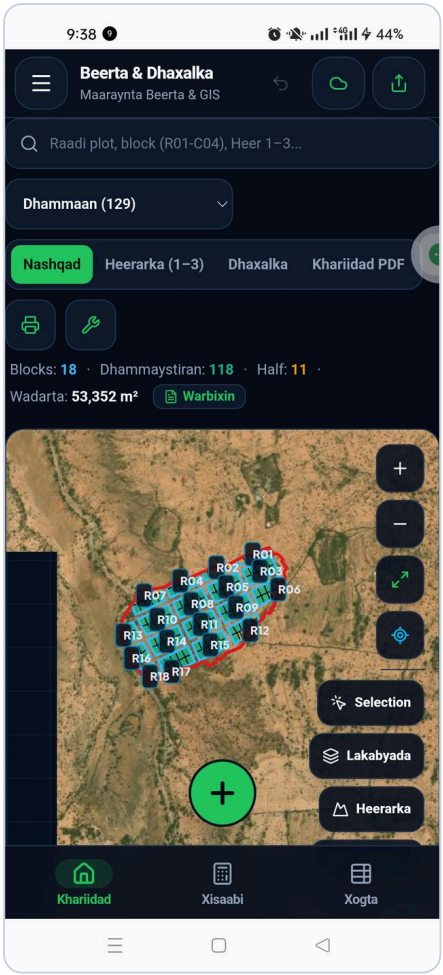
Shaashadaha 1–3: Bilow, Khariidad, Menu

Waxa uu qabto shaashad kasta



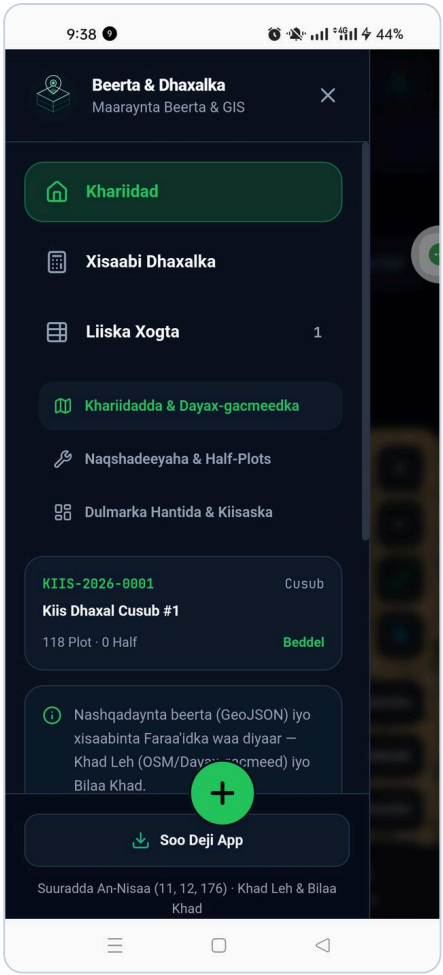
2 Bilow (splash)

Magaca appka, ujeedo gaaban, labo kaar (OSM & Dayax-gacmeed · Faraa'idka Islaamka "1 Kiis · 118 Plot oo diyaarsan"). "Bilow" waa CTA weyn; hoos waxaa ku qoran "Toos u fur Xisaabiyaha". Sare: xaaladda khadka + "Soo Deji App".



1 Khariidad (guri)

Raadin, filter (Dhammaan 129), 4 tab (Nashqad · Heerarka 1–3 · Dhaxalka · Khariidad PDF), print & wrench, tirada (18 block · 118 dhammaystiran · 11 half · 53,352 m²), khariidad dayax-gacmeed + badhamo (zoom, fullscreen, goobta, Selection, Lakabyada, Heerarka).



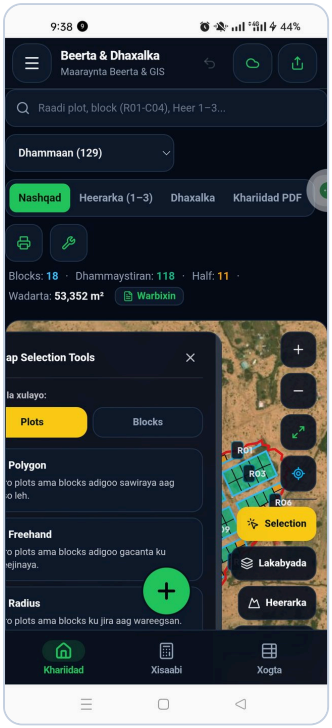
3 Menu bidix

Khariidad · Xisaabi Dhaxalka · Liiska Xogta (1) + 3 qaybood-hoosaad, kaarka kiiska hadda furan (118 Plot · 0 Half · Beddel), qoraal macluumaad, "Soo Deji App", iyo aayad-tixraac.

Fahamkayga: Bilowgu waa suuban yahay oo fariinta waa cad tahay. Khariidadda waa shaashadda hoyga appka, laakiin waxay isku daraysaa **filter, tabs, tools, stats iyo khariidad** hal shaashad — taasi waa halka dhibaatooyinka muuqaalka ugu badan ka bilowdaan (eeg bogga 8).

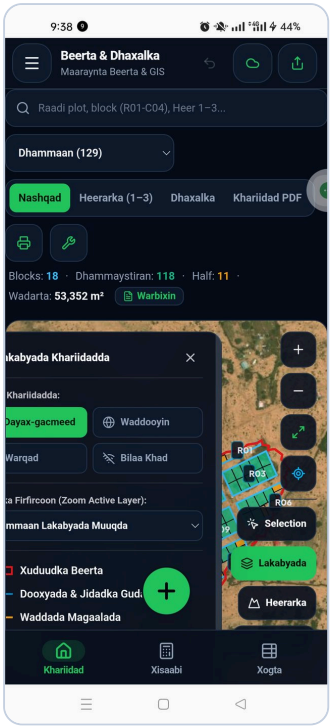
Shaashadaha 4–7: Qalabka khariidadda

Selection · Lakabyo · Heerarka · Menu-ga “+”



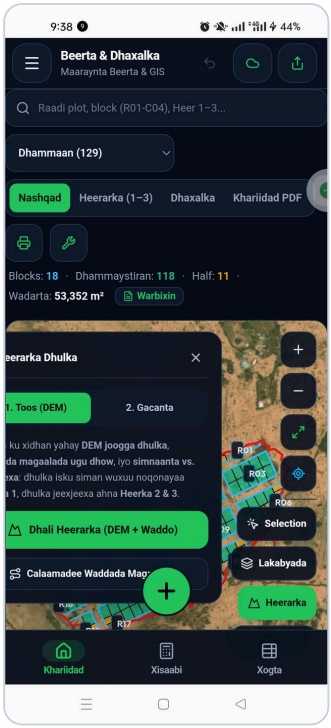
4 Selection

“Map Selection Tools”: la xulayo Plots ama Blocks; hab Polygon (sawir aag), Freehand (gacan-ka-sawir), Radius (goobo-wareeg).



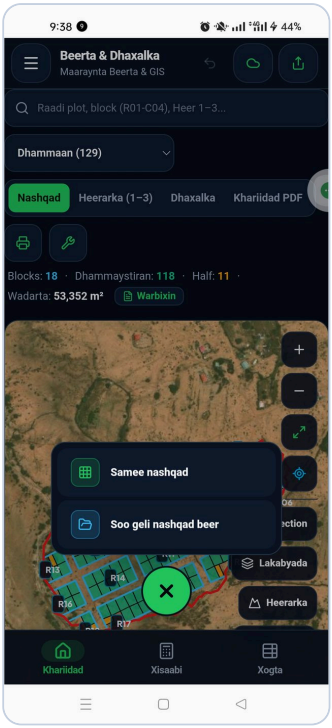
5 Lakabyada

Khariidado: Dayax-gacmeed · Waddooyin · Warqad · Bilaa Khad; “Zoom Active Layer”; legend (Xuduudka Beerta, Dooxyada & Jidadka, Waddada Magaalada).



6 Heerarka dhulka

Heerarka 1–3: “Toos (DEM)” ama “Gacanta”; DEM wuxuu cabbiraa joogga, dooxooyinka iyo simnaanta; badhan “Dhali Heerarka (DEM + Waddo)”.



7 Menu “+”

Labo ficil: “Samee nashqad” (naqshadee grid beer) iyo “Soo geli nashqad beer” (import).

Sida qalabka u shaqeeyo

Xulashada

Isticmaaluhu wuxuu doortaa nooca walaxda (Plots/Blocks) iyo qaabka xulashada. Natiijadu waxay u gudubtaa Xisaabiga ama Soo-saarka.

Lakabyada

Hal lakab oo hoose ayaa la shidaa; xuduudka beerta, jidadka iyo waddooyinka waxay yihiin lakabyo kor ka yaal.

Heerarka

Heer 1–3 waa fasalid dhul oo ku salaysan DEM; waxay saameyn ku leedahay qiimaynta plots-ka (waxaa la arkaa tab “Heerarka (1–3)”).

Muuqaal: shaashadaha 4, 5, 6 dhammaantood waxaa laga go’ay **dhinaca bidix** (cinwaannada “ap Selection Tools”, “akabyada”, “eerarka Dhulka” way go’an yihiin) — eeg khalad E1.

Shaashadaha 8–9: Xisaabi Dhaxalka

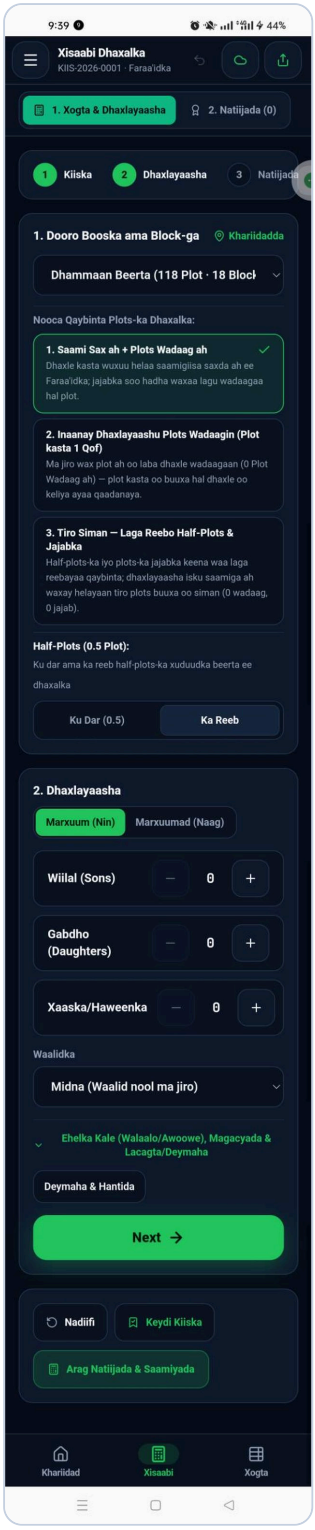
Wizard-ka xisaabinta: Kiiska → Dhaxlayaasha → Natiijada



8 Bilowga wizard-ka



Stepper-ka oo dhinaca ka go'an



9 Shaashad dheer oo dhammaystiran

1 · Kiiska

Dooro boos ama block ("Dhammaan Beerta · 118 Plot · 18 Block") iyo **nooca qaybinta**:

- **1. Saami sax ah + Plots wadaag ah** — dhaxle kasta wuxuu helaa saamigiisa saxda ah; jajabka ayaa lagu wadaagaa hal plot.
- **2. Plot kasta 1 qof** — ma jiraan plots la wadaago (0 wadaag); plot waxaa qaadanaya hal dhaxle.
- **3. Tiro siman** — half-plots iyo jajab waa laga reebay; dhaxlayaashu waxay helaan plots buuxa oo siman.

Half-Plots (0.5): "Ku dar / Ka reeb" — ma lagu xisaabiyaa half-plots xuduudka beerta.

2 · Dhaxlayaasha

- Marxuum (Nin) / Marxuumad (Naag)
- Counters: Wiilal · Gabdho · Xaaska/Haweenka
- Waalidka (dropdown: "Midna — waalid nool ma jiro")
- Ehelka kale: walaalo/awoowe, magacyo & lacag
- "Deymaha & Hantida" (chip)

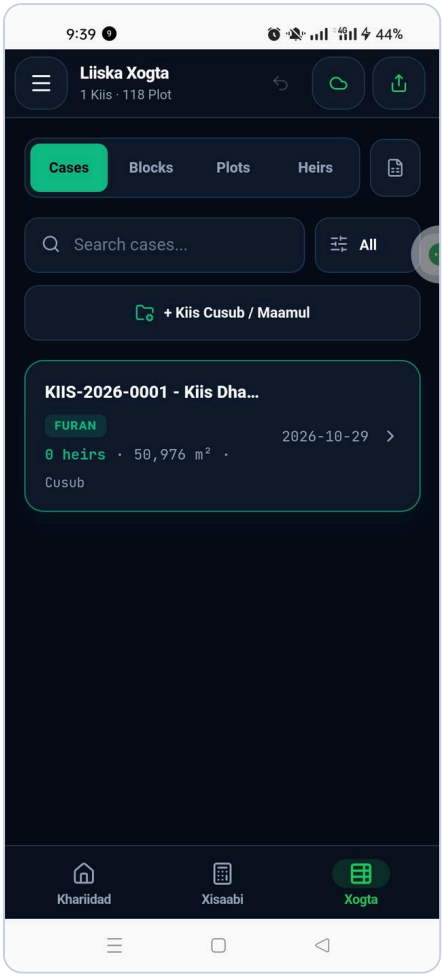
3 · Natiijada

Badhamada: **Next →**, **Nadiifi**, **Keydi Kiiska**, **Arag Natiijada & Saamiyada**. Tab-ka 2 wuxuu muujiyaa tirada natiijooyinka "(0)".

Maxaan ka fahmay: marka dhaxlayaasha la geliyo, appku wuxuu saami kasta u beddelaa tiro plots ah, adigoo raacaya nooca qaybinta ee la doortay. Natiijada lama arkin shaashadaha, sidaas darteed ma xaqiijin karo saxnaanta xisaabta.

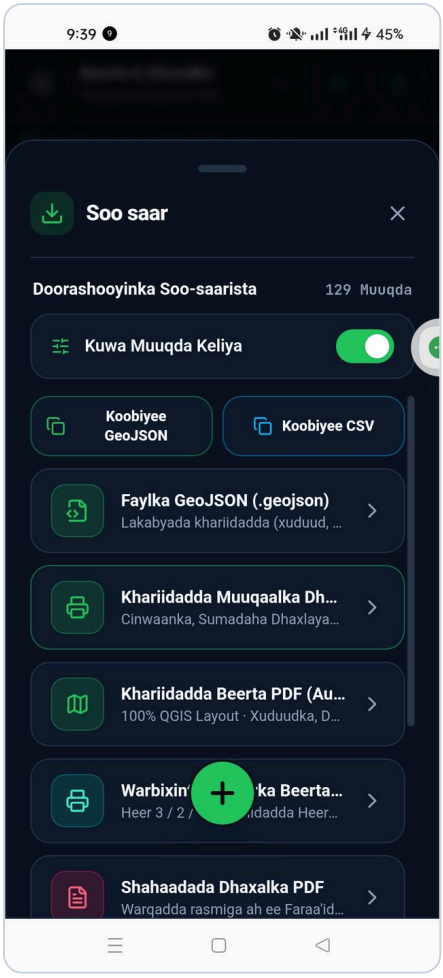
Shaashadaha 10–12: Xogta, Soo saar, Layout

Diiwaanka, dukumeentiyada iyo daabacaaddа



10 **Liiska Xogta**

4 tab: Cases · Blocks · Plots · Heirs; raadin + filter; "+ Kiis Cusub / Maamul"; kaar kiis leh **FURAN**, 0 heirs, 50,976 m², "Cusub", taariikh 2026-10-29.



11 **Soo saar**

"Soo saar" bottom-sheet: 129 muuqda; toggle "Kuwa Muuqda Keliya"; koobiyee GeoJSON/CSV; faylka GeoJSON; khariidad muuqaal; Khariidadda Beerta PDF (100% QGIS layout); warbixin; Shahaadada Dhaxalka PDF.



12 **Layout (PDF preview)**

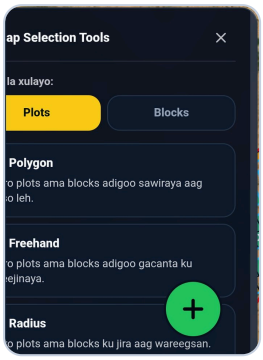
Tabs: Khariidadda Dhaxalka · Layout 11 · Warbixin; A4 Landscape (Ballac); qaab "Tiirka Dhinaca". Bogga wuxuu leeyahay khariidad, Waqooyi, cabir 1:2,750, legend, xog-ilaha (WGS84/UTM 38N) iyo kiis ID.

Qiimaha: layout-ku wuu xirfad badan yahay — cabir, grid-ka isku-xirka (UTM), legend iyo xog-ilaha. Waa tilmaame cad oo dukumeenti rasmi ah (Notary/Land Office) laga dhigi karo.

Dhibaato: preview-ga aad buu yar yahay (bogga wuxuu qabtaa ≈ 1/4 shaashadda) oo meel madhan oo badan ayaa ku hareeraysan; magacyada ka soo saar way go'an yihiin ("Khariidadda Muuqaalka Dh...").

Khaladaadka (1): Muuqaal & layout

E1–E5 oo leh caddayn ka timid shaashadaha dhabta ah



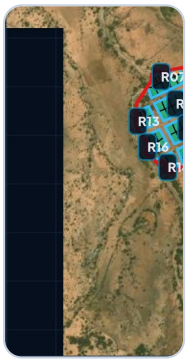
E1 · Panels-ka khariidadda waa bidix ka go’an Sare

Selection, Lakabyada iyo Heerarka waxay ka soo baxaan dhinaca bidix oo waxaa ka go’an xarfaha hore (“ap Selection Tools”, “la xulayo”). Panel-ku wuu ka baxsan yahay shaashadda — isticmaaluhu ma arki karo qayb ka mid ah badhamada iyo qoraalka.



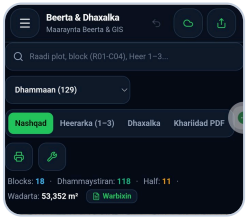
E2 · Badhanka “+” (FAB) wuxuu qarinayaa xogta Sare

FAB waxa uu ku fadhiyaa dhexda hoose ee ku dhawaad shaashad kasta: wuxuu ku dul-yaal ka soo-saarka (“Warbixin... Beerta”), menu-ga bidix, iyo panels-ka — wuxuu daboolaa qoraal iyo badhamo la riixi karo.



E3 · Rectangle madow oo bidix ka yaal khariidadda Dhexe

Waxaa khariidadda bidixdeeda ka muuqda goob madow oo leh xariijimo — waxay u egtahay tile aan soo dhicin ama lakab madhan. Waxay u muuqataa jajab ama khalad.



E4 · Khariidaddu waa yartahay: ≈ 36% oo keliya shaashadda Dhexe

Header + raadin + dropdown + 4 tab + labo icon + stats ayaa qaadanaya ≈ 37% dhanka sare, bar-ka hoose 8%; khariidadda lafteeda waxaa u hadhay ≈ 36% (860px ka mid ah 2400px). Marka panel la furo, khariidaddu way dhammaanaysaa.

E5 · Calaamadaha R01…R18 waa waaweyn yihiin oo is-dul-fadhiya Dhexe

Calaamad kasta waa kaar madow oo weyn; R01/R03, R17/R18 iyo R12/R15 way is-dul-yaalaan, waxayna qarinayaan plots-ka hoostooda. Waa in la yareeyo oo la helo “collision hiding” (dhowr calaamad xad-dhaaf ah oo la qariyo marka zoom yar yahay).

Khaladaadka (2): Xog & is-waafaqsanaan

Tirooyinka isku mid ma aha shaashad kasta

E10 · Saddex qiyaasood oo dhul ah Sare

Shaashad	Qiimaha	Magaca
#1 Khariidad	**53,352 m²**	“Wadarta”
#10 Liiska Xogta	**50,976 m²**	(magac la’aan, kiis)
#12 Layout PDF	**9.57 ha = 95,700 m²**	“Xuduudka Rasmiga ah ee Beerta”

Waxaa suurtagal ah inay yihiin saddex wax oo kala duwan (xuduudka, wadarta plots, plots-ka kiiska), laakiin **midna ma sheegayo waxa uu yahay**. Dukumeenti rasmi ah (shahaado) hal tiro oo cad ayuu u baahan yahay.

E11 · Tirada plots-ka Sare

- Khariidad: “Dhammaystiran 118” · “Half 11” · filter “129”.
- Menu: “118 Plot · **0 Half**” (halka khariidaddu tiraysay 11).
- Layout: “Plots: **129**” iyo legend “118 Plot / 11 Half”.
- Soo saar: “129 Muuqda”; Xogta: “118 Plot”.

129 = 118 + 11 waa macquul, laakiin magacyada “Plot”, “Dhammaystiran”, “Muuqda” way is-beddelayaan.

E12 · Kiis “FURAN” oo 0 dhaxle leh Hoose

Kaarka kiiska wuxuu muujiyaa “0 heirs” iyo xaalad FURAN; tab-ka natiijada “(0)”. Waa xaalad caadi ah oo kiis cusub, laakiin xaaladda madhan looma muujin si cad (empty state).

E13 · “Arag Natiijada” iyada oo xog la’aan ah Dhexe

Badhanku waa shidan yahay iyadoo Wiilal/Gabdho/Xaaska dhammaantood 0 yihiin. Ma garanayo waxa dhacaya marka la riixo; waa in la muujiyo fariin (“Ugu yaraan hal dhaxle geli”) ama la damiyo.

E14 · Saddex hab oo loo gudbo natiijada Hoose

“Next →”, “Arag Natiijada & Saamiyada” iyo tab “2. Natiijada (0)” oo dhammaantood ku aadaya meel isku mid ah — isticmaalaha way ku walwalsan yihiin.

E6 · Stepper-ku wuu is-diidan yahay Dhexe



Tallaabo 1 iyo 2 labadaba cagaar buuxa ayay yihiin (“la dhammeeyay”), halka tallaabo 2 tahay shaashadda hadda taal; tallaabo 3 (“Natiijada”) waa go’antahay. Midab uun looma kala saari karo “la dhammeeyay” iyo “hadda”.

Khaladaadka (3): Luuqad, isticmaal & kooban

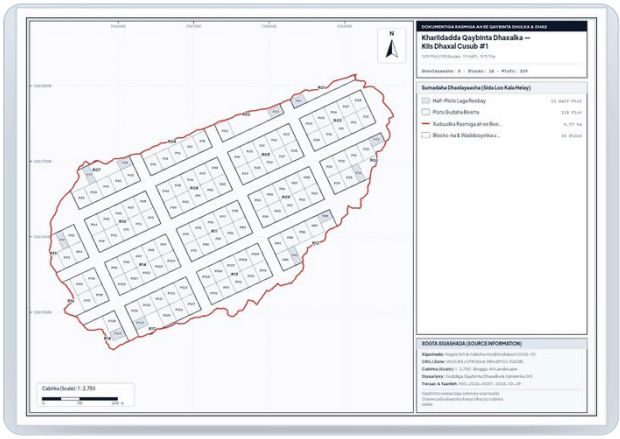
Soo koobid dhammaan 18-ka khalad iyo mudnaanta

E7 · Qoraal go’an Dhexe



“KIIS-2026-0001 - Kiis Dha...”, “Dhammaan Beerta (118 Plot · 18 Block” oo go’an, “Khariidadda Muuqaalka Dh...”. Magacyada muhiimka ah waa in la duubo 2 line.

E8 · Preview-ga layout Dhexe



Bogga PDF-ka wuxuu muuqdaa yar, dhexda shaashadda; saqafka iyo hoosta waa madhan yihiin. Ma jiro zoom ama full-screen.

E15 · Luuqad isku dhafan Dhexe

Ingiriis: *Selection, Blocks, Plots, Cases, Heirs, Next, Map Selection Tools, Half, Polygon, Freehand, Radius*. Soomaali: Kiis, Dhaxlayaasha, Khariidad. Isla fikrad ahaan: “Kiis / Case”, “Dhaxlayaal / Heirs”, “Plot / Boos”.

E16 · Aqoonsi & aqoon-yar Dhexe

Qoraal sharaxaad (cawl-guduud ah) oo yar ~11–12px ah oo ku yaal asal madow-buluug ah; dadka da’da ah iyo shaashadaha dayaxa-goobta waa kuwo aqoon-yar.

E17 · Icons aan magac lahayn Dhexe

↶, ☁, ⇪, 🖨 iyo 🔧 ma laha qoraal; “Naqshadeeyaha & Half-Plots” (menu) iyo wrench (sare) waxay u egyihiin isla shaqo.

E9 / E18 Hoose

Tab-ka “Khariidad PDF” wuu go’an yahay (scroll jiheyn ma leh). “Soo geli nashqad beer” waa eray aan caddayn.

Liiska dhammaystiran ee mudnaanta

ID	Khalad	Culeys
E1	Panels bidix ka go’an	Sare
E2	FAB wuxuu qarinayaa xogta	Sare
E10	3 qiyaas oo dhul ah	Sare
E11	Tirooyinka plots isku mid ma aha	Sare
E3	Rectangle madow khariidadda	Dhexe
E4	Khariidadda 36% oo keliya	Dhexe
E5	Calaamadaha is-dul-fadhiya	Dhexe
E6	Stepper is-diidan	Dhexe
E7	Qoraal go’an (truncate)	Dhexe

ID	Khalad	Culeys
E8	Preview-ga layout aad u yar	Dhexe
E13	Natiijo iyadoo xog la’aan ah	Dhexe
E15	Soomaali + Ingiriis isku dhafan	Dhexe
E16	Qoraal yar & muted (aqoon-yar)	Dhexe
E17	Icons aan magac lahayn	Dhexe
E9	Tab-ka sare oo scroll ah ayan tilmaan lahayn	Hoose
E12	Empty state ma jiro	Hoose
E14	3 hab oo loo gudbo natiijada	Hoose
E18	Menu-ga “+” oo erayo qaboojiya	Hoose

Sare 4 · Dhexe 10 · Hoose 4 = 18. Qoraalka ugu horreeya waa waxa aan arkay; saxnaanta xisaabinta dhaxalka lama xaqiijin sababtoo ah natiijada lama arkin.

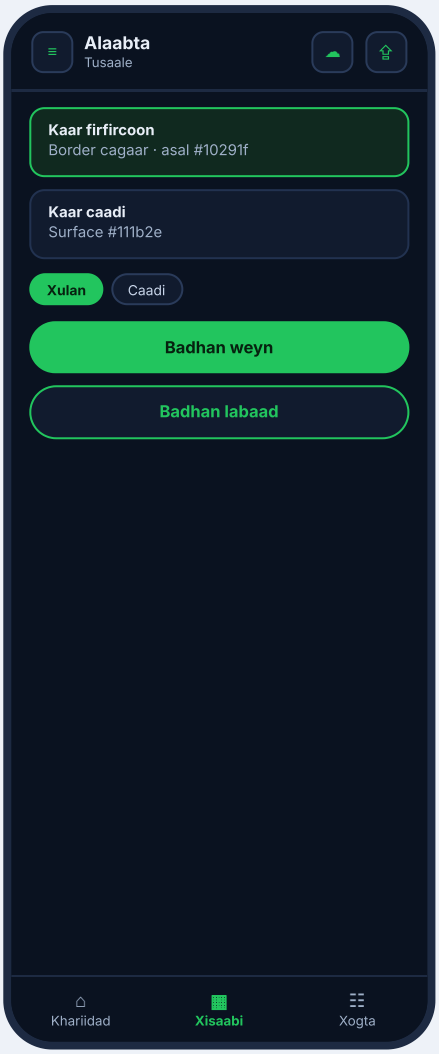
Jihada design-ka: nidaam midab, far & cabbir

Design tokens oo la isticmaali karo shaashad kasta

1 · Midabada (waxay ku salaysan yihiin kuwa hadda jira)

bg-0 · Asal #060d18	**bg-1 · App** #0a1220	**Surface · Kaar** #111b2e
Border #223250	**Primary · Cagaar** #22c55e	**Primary-dark** #16a34a
Info · Buluug #38bdf8	**Xulasho · Jaalle** #facc15	**Khalad** #f87171
Text #e8eef7	**Muted (cusub)** #a9b8d0	

Isbeddel keliya: **Muted** #8a9bb4 → #a9b8d0 si contrast-ku u gaaro ≥ 4.5:1 xagga qoraalka yar.



2 · Far (Inter / Roboto)

Cinwaan 22/700

Qaybta 18/600

Qoraal 16/400 — ugu yaraan 14

Sharaxaad 13/400 · ugu yar 12 — marna 11 ha ka yaraan

3 · Cabbirka & geesaha



Spacing: 4 · 8 · 12 · 16 · 24 · 32



Radius: 8 · 12 · 16 · 24 · Touch target ≥ 48dp

4 · Shanta mabda' oo la raaco

Khariidad hore Khariidaddu ugu yaraan 60% shaashadda	**Hal tiro, hal magac** Tiro kasta hal meel ayaa laga xisaabiyaa	**Sheet, ma panel** Qalab kasta bottom-sheet buuxa	**Soomaali buuxda** Erayo Ingiriis ah waa la beddelaa	**Hal ficil weyn** Shaashad kasta hal CTA oo cagaar ah

Qaab-dhismeedka macluumaadka (Sitemap)

Ka dhig 4 tab oo cad — ka saar isku-celinta menu-ga, bar-ka iyo FAB

Beerta & Dhaxalka
App shell · header ≡ Kiiska hadda furan · ☁ Keydin · ⚙ Fadhi

① Khariidad

- Raadin (floating)
- Nashqad: samee / soo geli
- Xulasho: Polygon · Freehand · Radius
- Lakabyada & legend
- Heerarka (DEM · Gacanta)
- Naqshadeeyaha & Half-Plots

② Xisaabi

- Tallaabo 1 · Kiiska
- Tallaabo 2 · Dhaxlayaasha
- Tallaabo 3 · Natiijada
- Keydi kiis
- Nadiifi

③ Xogta

- Cases · Blocks · Plots · Heirs
- Raadin & filter
- Kiis cusub / Maamul
- Dulmarka Hantida & Kiisaska

④ Soo saar

- Xog: GeoJSON · CSV
- Khariidad: PDF · muuqaal
- Layout designer
- Warbixin
- Shahaadada Dhaxalka PDF

Hadda (qaabka jira)

- 3 tab hoose + menu bidix + FAB “+” + 4 tab sare + icon-yo sare — **5 meelood** ayaa fidiyey isla shaqooyinka.
- “Soo saar” waa icon sare (⇪) iyo tab “Khariidad PDF” iyo menu “Soo Deji”.
- “Liiska Xogta” (menu) = “Xogta” (tab).

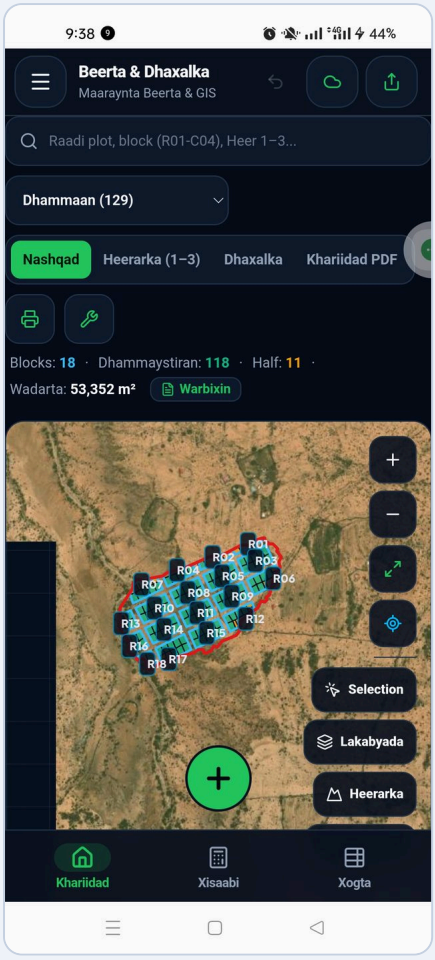
Hadda la soo jeediyey

- **4 tab hoose** oo magac leh: Khariidad · Xisaabi · Xogta · Soo saar.
- Menu-ga bidix wuxuu noqonayaa **Fadhi** (luuqad, offline, ku saabsan, Suuradda).
- Tab-yada sare ee khariidadda waxay noqdaan **qalab ku jira sheet-ka**.
- FAB “+” wuxuu ku dhex jiraa tab-ka Khariidad oo keliya.

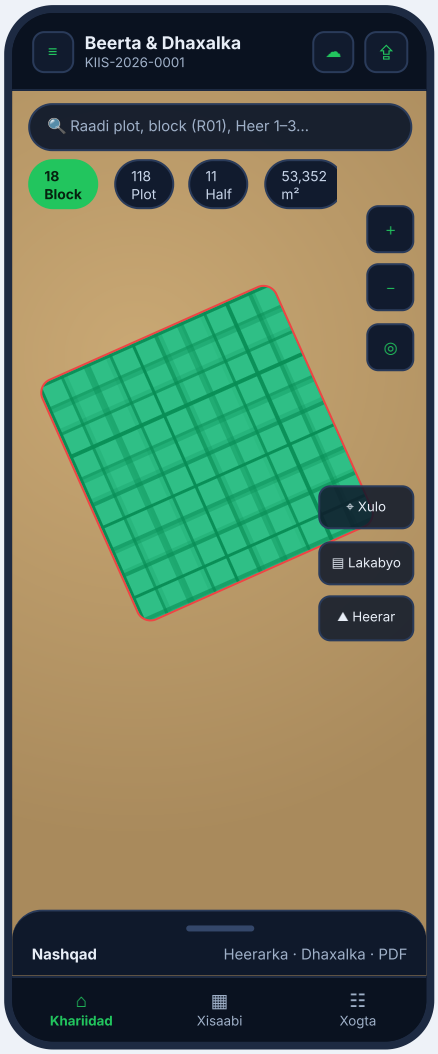
Xusuusin: dhammaan shaqooyinka jira waa la kaydinayaa — waxaa isbeddelaya **halka ay ku yaalaan** iyo sida loo gaaro.

Khariidadda: hore iyo hadda

Khariidad weyn, tools oo magac leh, sheet hoose — shaqada isku mid



Hadda — khariidad 36%



Soo jeedin — khariidad ≈ 68%

Isbeddellada

1. Raadinta waxay noqotaa pill ku sabbaysan khariidadda.
2. Stats-ka waxay noqdaan chips hal saf ah (scroll) — halka 2 saf.
3. Zoom/goobta waxay isku xirnaadaan rail yar.
4. Xulo · Lakabyo · Heerar: badhamo **magac leh** oo ku sii jira rail hoose.
5. Tab-yada (Nashqad/Heerarka/Dhaxalka/PDF) waxay ku wareegaan sheet hoose oo la jiidi karo.
6. Bar hoose 4 tab (eeg bogga 12).

Calaamadaha R01...R18

Qaab yar oo xarun-ka plot-ka saaran; marka zoom < 17 waxaa la muujiyaa block-yada oo keliya; marka zoom ≥ 18 ayaa la muujiyaa P-numbers. Tani waxay meesha ka saaraysaa E5.

Goobta madow (E3)

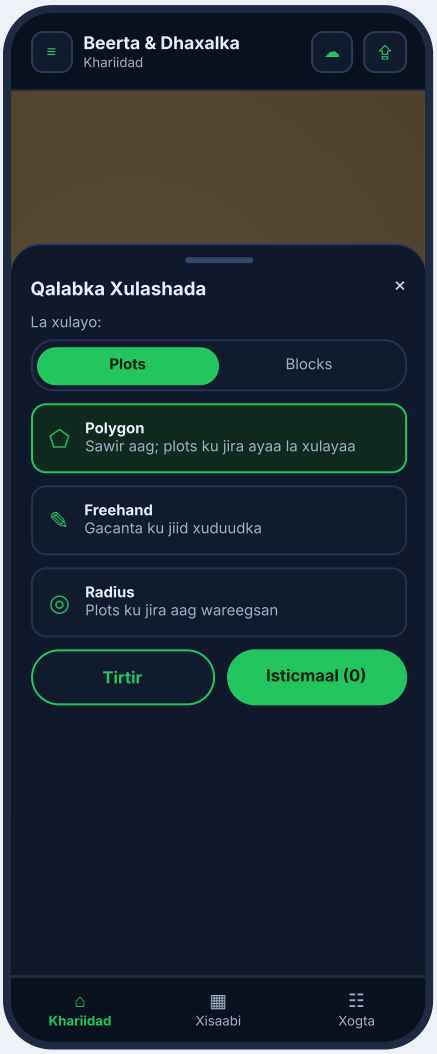
Asalka khariidadda u dhig midab dhexdhexaad ah (#a98a5c) ilaa tile-ku soo dhaco; ha muujin lakab madhan.

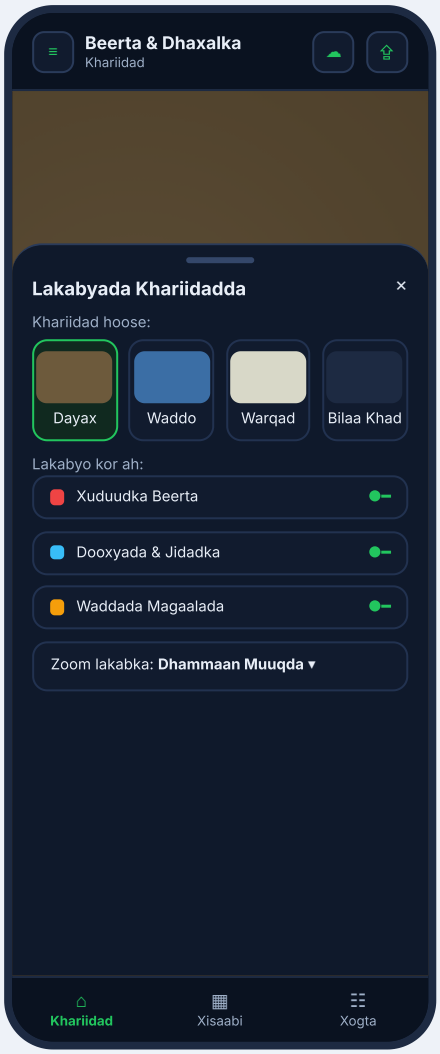
Xaalado

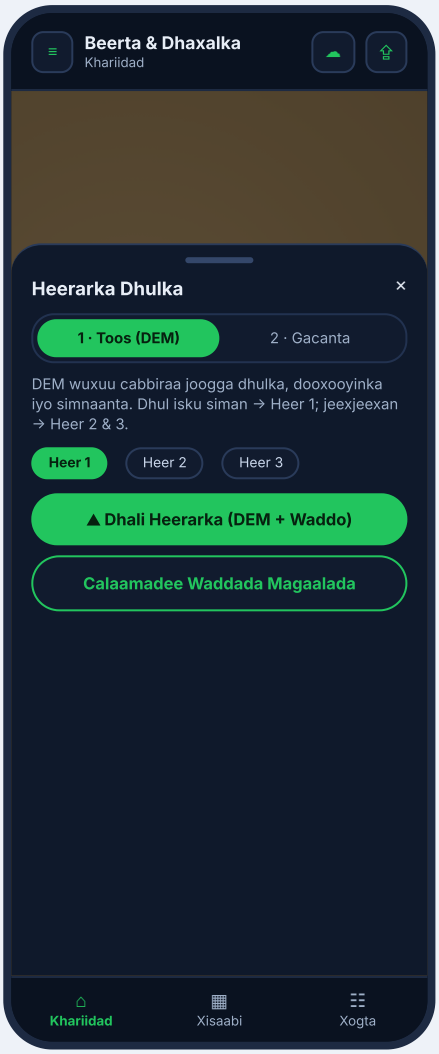
Loading: skeleton cagaar ah. Offline: calaamad “Khad Leh” ayaa ku jirta chip-ka sare.

Qalabka khariidadda: bottom-sheet buuxa

Selection · Lakabyo · Heerar — wax go’an ma jiraan, FAB-na wuu dhumaa







Xalka E1 iyo E2

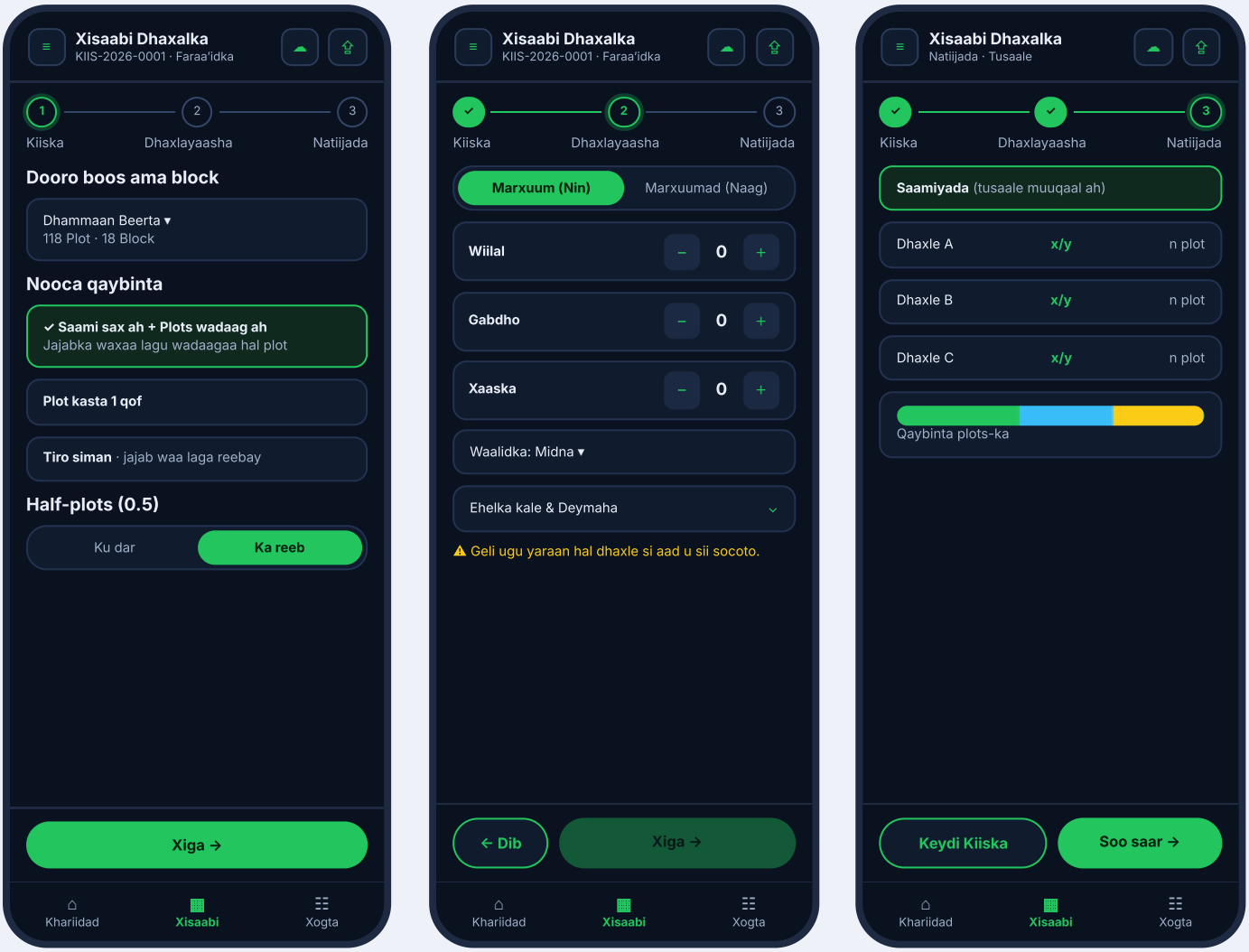
1. Sheet-ku wuxuu buuxiyaa ballaca shaashadda 100% — qoraal lagama go’o.
2. Cinwaannada oo Soomaali buuxa ah: “Qalabka Xulashada” (ma aha “Map Selection Tools”).
3. Marka sheet furan yahay, FAB wuu dhumaa; badhamaha ficilka waxay ku jiraan sheet-ka gudihiisa.
4. Sheet wuxuu xirmaa ×, jiid hoos, ama riix khariidadda.

Isku-mid-ka shaqada

Habka Polygon/Freehand/Radius, lakabyada iyo DEM waa isku mid — kaliya meesha loogu soo bandhigo ayaa isbeddelaysa. “Isticmaal (0)” wuxuu muujiyaa tirada la xulay, markaas isticmaaluhu wuu og yahay waxa uu samaynayo.

Xisaabi Dhaxalka: wizard cad

Stepper saxan, CTA hal ah, ka hortag khalad



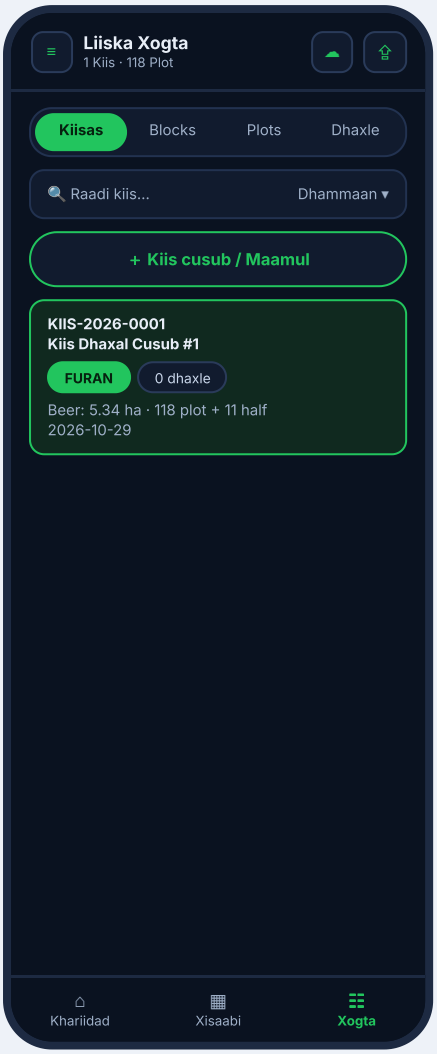
Isbeddellada

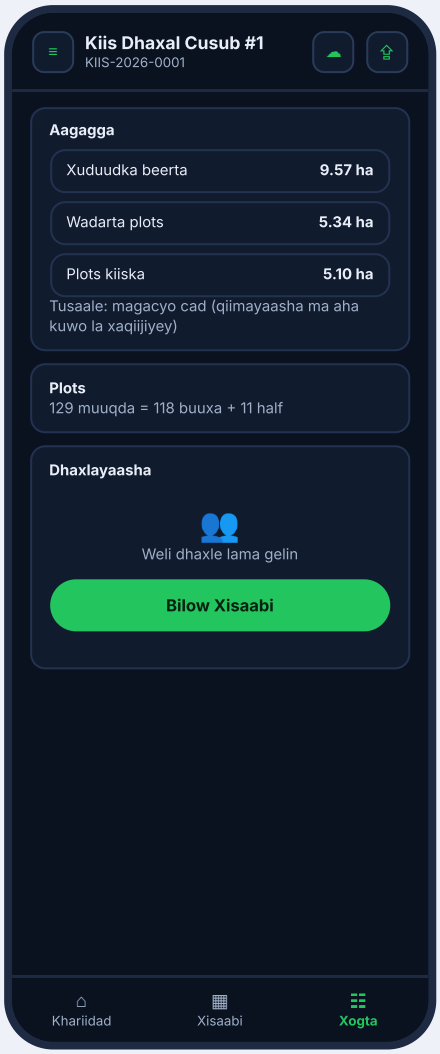
1. **Stepper:** ✓ = la dhammeeyey · cagaar xariijin = hadda · madow = soo socda (xalka E6).
2. **Footer joogto ah:** Dib / Xiga — wuxuu beddelayaa “Next”, “Arag Natiijada” iyo tab-ka natiijada (E14).
3. **Ka hortag:** “Xiga” waa dami ilaa ugu yaraan hal dhaxle la geliyo; fariin jaalle ah ayaa sharaxaysa (E13).
4. **Counters:** badhamaha − + 48dp; ereyada Soomaali ah (“Wiilal”, “Gabdho”) iyo Ingiriis la gooyay.

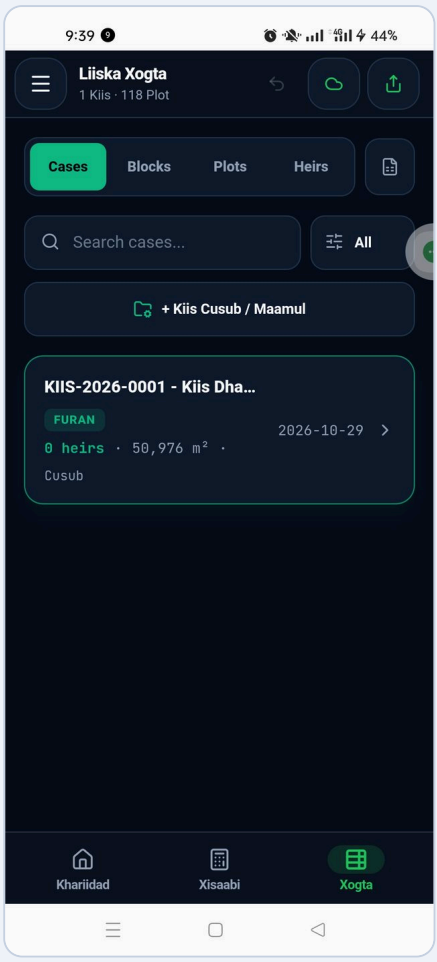
Digniin: shaashadda 3 waa tusaale muuqaal ah. Natiijada dhabta ah ee appku bixiyo ma aan arkin, sidaas darteed tirooyinka (x/y, n plot) waa meel-buuxiye; qaabka xisaabta ma aan taaban.

Xogta: magac cad, tiro cad, xaalad madhan

Xalka E7, E10, E11 iyo E12







Hadda — cinwaan go'an, "0 heirs", 50,976 m² (magac la'aan)

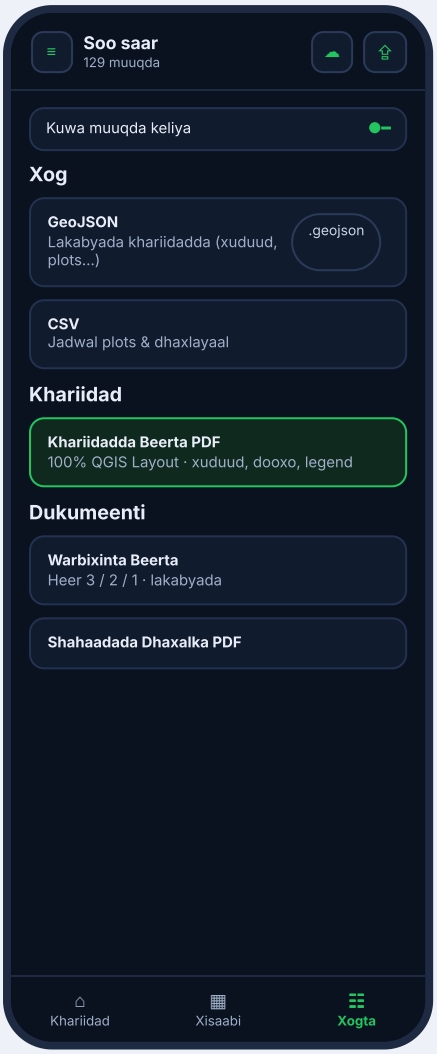
Isbeddellada

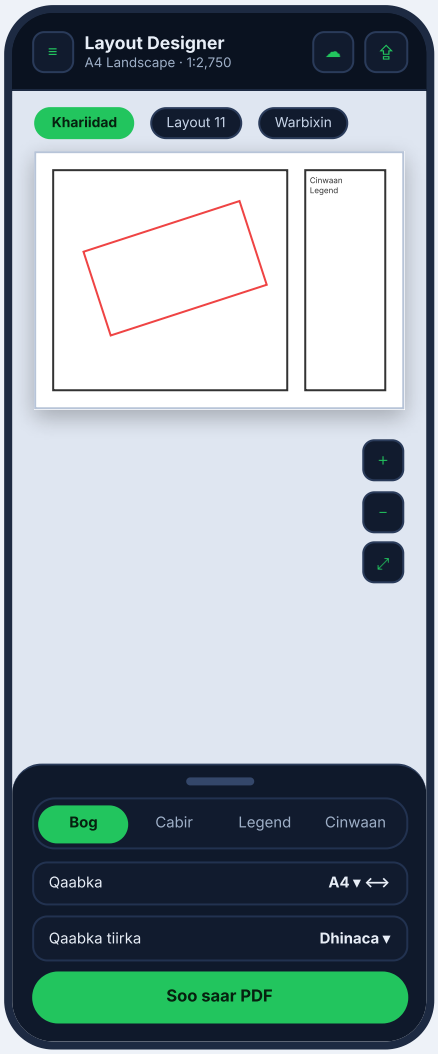
1. Cinwaanka kiiska wuxuu u duubmaa **2 saf** halkii la gooyi lahaa (E7).
2. Aag kasta wuxuu leeyahay **magac cad**: Xuduudka · Wadarta plots · Plots-ka kiiska (E10).
3. Hal "Xisaabiye Xog" (single source of truth) ayaa tirinaya plots, half iyo m² — shaashad kasta isagay akhrisaa (E11).
4. Empty state: sawir + erey + badhan "Bilow Xisaabi" (E12).
5. Chip-yada xaaladda: FURAN · DHAMMAAD · KEYDIYAY.

Muhiim: haddii saddexda qiyaasood ay dhab ahaan yihiin wax kala duwan, appku waa saxan yahay — kaliya magacyadu waa maqan yihiin. Haddii ay isku shay yihiin oo kala duwan, waa **bug xisaabeed** oo lagu baadho koodhka (eeg bogga 18). Qiimayaasha sawirka waa tusaale muuqaal ah.

Soo saar & Layout tools

Kooxeyn xog/khariidad/dukumeenti; preview weyn oo zoom leh

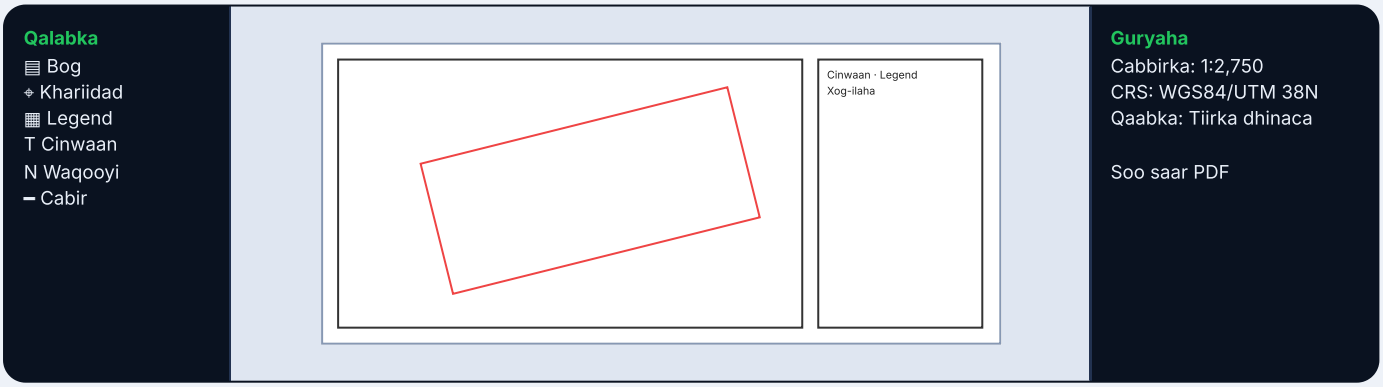




Layout tools (sida QGIS)

1. **Bog:** A4 / A3 · Landscape / Portrait.
2. **Cabir:** 1:2,750 oo la beddeli karo; bar-cabir toos ah.
3. **Legend:** shid/dami shay kasta.
4. **Cinwaan & xog-ilaha:** CRS, taariikh, ID kiis.
5. **Waqooyi (N)** iyo grid UTM.
6. **Preview:** pinch-zoom, full-screen (E8).

Muuqaalka tablet/desktop: Layout Designer oo saddex-qaybood ah



Architecture-ka la soo jeediyey

Qaybo cad: UI → Store → Domain → Services → Platform

Tani waa jaangoyn la soo jeediyey oo ku salaysan shaashadaha. Koodhka dhabta ah lama arkin; qaybaha jira waa la dhaqaajiyaa (move) — **ma beddelayo xisaabta**, kaliya meesha ay ku nooshahay.

① Presentation (UI)	Map screen	Xisaabi wizard	Xogta list	Soo saar & Layout	Fadhi
② State (Store)	caseStore kiis hadda furan	parcelStore blocks / plots	heirsStore dhaxlayaal	uiStore sheet, tab, step	
③ Domain (logic)	FaraidEngine saamiyada	ParcelAllocator plots → dhaxle	Metrics m², plots, half (hal meel)	Validators ka hortag	
④ Services	GeoService tiles, DEM, xulasho	StorageService IndexedDB	ExportService GeoJSON · CSV · PDF	SyncService ☁ keydin	
⑤ Platform	PWA / Service Worker (offline)	APK wrapper	OSM · dayax-gacmeed tiles	Cache khariidad	

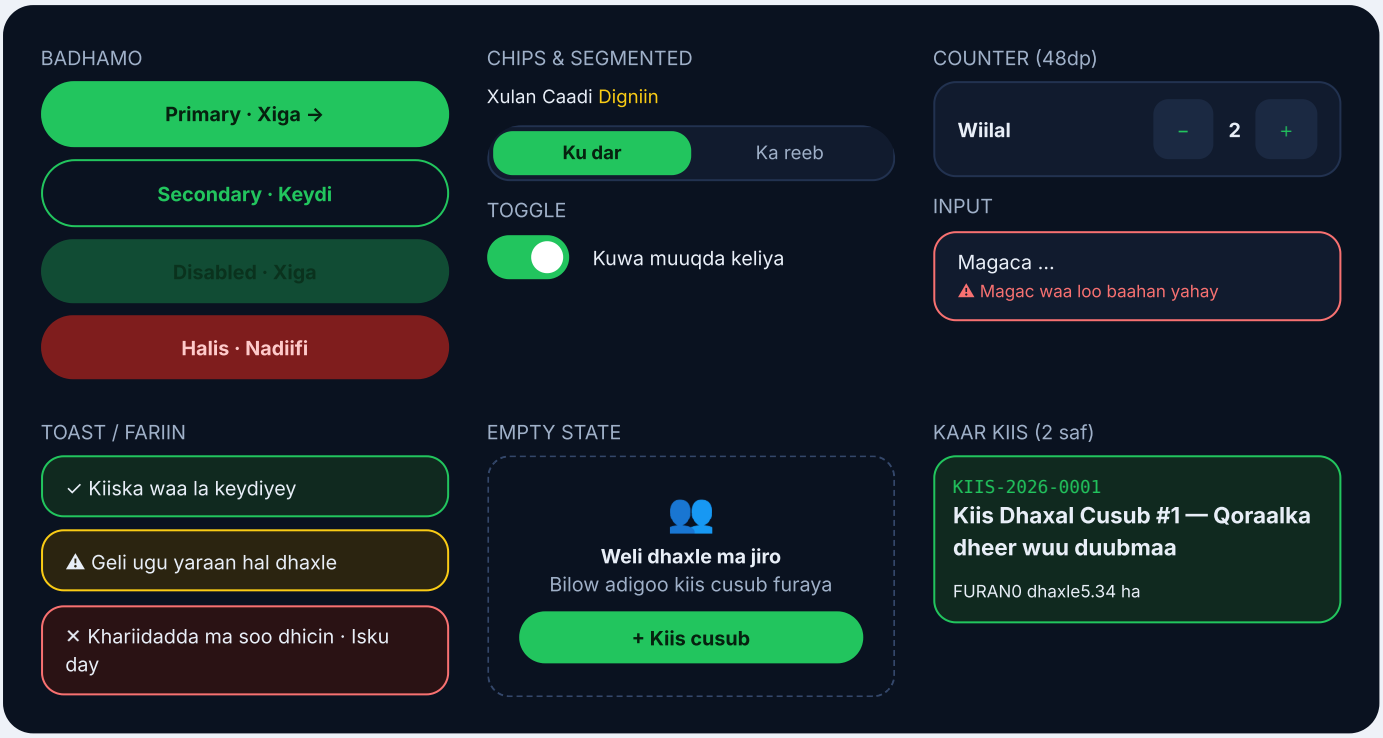
```
/src
 /ui        screens/ sheets/ components/
 /state     caseStore parcelStore uiStore
 /domain    faraid/ allocation/ metrics/
 /services  geo/ storage/ export/ sync/
 /i18n      so.json   (Soomaali — hal meel)
 /theme     tokens.css (bogga 11)
```

Faa'iidooyinka

- **Metrics** hal meel = E10 & E11 meesha ka baxaan.
- **/i18n** hal fayl = E15 (luuqad isku dhafan) la xalliyo.
- **Tokens** = muuqaal isku mid ah.
- Domain wuu tijaabin karaa oo UI laga' madax-bannaan.

Maktabadda qaybaha (components) & xaaladaha

Qayb kasta: caadi · firfircoon · dami · khalad



Xeerarka qaybaha

- Ugu yaraan **48dp** goob la taabto.
- Qoraal ≥ **12px**; qoraal muuqda ≥ 4.5:1 contrast.
- Icon kasta + **qoraal** (ama tooltip) — marna icon keliya.
- Cinwaan muhiim ah → duub 2 saf, ha gooyin.

Qaybaha lagu daro (cusub)

- **BottomSheet** — qalab kasta (bogga 14).
- **Stepper** 3 xaalad (bogga 15).
- **EmptyState** & **Skeleton**.
- **Toast** — keydin, khalad, offline.

Hawl-gal, responsive & talaabooyinka xiga

Sida loo hirgeliyo adigoo shaqada aan taabanayn

Responsive: hal appka, saddex cabbir



Mobile · bar hoose + sheet

Tablet · rail bidix + sheet dhinaca

Desktop · sidebar + khariidad + inspector

Jadwalka hirgelinta

Wejiga	Waxa la sameeyo	Khaladaadka la xalliyo
1 · Dhaqso 1–3 maalmood	Panels bidix → sheet buuxa; FAB la qari marka sheet furan; stepper xaaladaha saxda; qoraal 2 saf; asal khariidad.	E1 E2 E3 E6 E7 E9
2 · Xog & luuqad 3–5 maalmood	Metrics hal meel; magacyada aagagga; Soomaali buuxa (i18n); empty states; validation “Xiga”.	E10 E11 E12 E13 E14 E15 E18
3 · Qaab cusub 1–2 toddobaad	4 tab; sheet-ka khariidadda; calaamadaha adaptive; layout preview zoom; tokens & components.	E4 E5 E8 E16 E17

Liiska hubinta kahor sii-deynta

- ☐ Dhammaan sheets-ka waxay buuxiyaan ballaca shaashadda 360–430px.
- ☐ Tirooyinka beerta waxay isku mid yihiin shaashad kasta (Khariidad · Xogta · Layout · Soo saar).
- ☐ Natiijada xisaabta dhaxalka way sidaanba tahay (tijaabo isbarbar dhig ah kahor iyo kadib).
- ☐ Ma jiraan erayo Ingiriis ah oo aan loo baahnayn.
- ☐ Icon kasta wuxuu leeyahay magac ama tooltip.
- ☐ Offline: khariidad, keydin iyo soo-saar way shaqeeyaan.

Talaabada xigta: haddii aad rabto, waxaan u beddeli karaa bogagga 13–17 mockup HTML/CSS oo toos loogu isticmaali karo appkaaga (token CSS + components), adigoo isticmaalaya fayl-ka appka haddii aad i soo gudbiso.